A DATA-DRIVEN INTERNSHIP MATCHING SYSTEM FOR SRI LANKAN STUDENTS USING WEB SCRAPING, NLP, AND COSINE SIMILARITY

BY

Ajithkumar A F A

BICT Honours Degree **2026**

A DATA-DRIVEN INTERNSHIP MATCHING SYSTEM FOR SRI LANKAN STUDENTS USING WEB SCRAPING, NLP, AND COSINE SIMILARITY

01. Ajithkumar A F A – ICT/21/804

Multimedia Technology

Department of Information and Communication Technology

Faculty of Technology

University of Sri Jayewardenepura

ict21804@fot.sjp.ac.lk / 076 210 8502

Dr. Pulasthi Gunawardhana (Main Supervisor)

Lecturer, Faculty of Technology, USJ

pulasthi@sjp.ac.lk

Mr. Akalanka Pnnipitiya (Co-supervisor)

Lecturer, Faculty of Technology, USJ

akalankap@sjp.ac.lk

Declaration

I certify that this thesis does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due reference is made in the text.

Group Members: Ajithkumar A F A

Supervisor : Dr. Pulasthi Gunawardhana (Main Supervisor)

(Lecturer, Faculty of Technology, USJ)

Mr Akalanka Pnnipitiya (Co-supervisor)

(Lecturer, Faculty of Technology, USJ

Acknowledgment

This thesis is prepared as an initial requirement for the final year research project which is a part of the Bachelor of Information and Communication Technology Honours Degree program offered by the Department of Information and Communication Technology, Faculty of Technology, University of Sri Jayewardenepura. The final year group research project is a milestone in the degree program that should be completed with the highest contribution of the group members and the support and guidance from many resource persons. Therefore, this page of the project thesis is reserved to acknowledge those valuable people.

First of all, we would like to express our deepest gratitude to our main supervisor, Dr Pulasthi Gunawardhana (Lecturer, Faculty of Technology, USJ) for filling us with their knowledge and experience of academic research and co-supervisor Mr Akalanka Pannipitiya (Lecturer, Faculty of Technology, USJ). As beginners for research projects, our biggest strength is having supportive and experienced supervisors and their guidance.

And also, it is necessary to acknowledge Dr Nalaka Lankasena (Head of the Department - Department of ICT, Faculty of Technology, USJ), for teaching us about literature study and academic writing and Mr Uthpala Samarathunga (Lecturer, Faculty of Technology, USJ) and all the other lecturers of Faculty of Technology, USJ, who helped us in every possible way. Behind each step of our group project, the Project Administration and Evaluation Committee (PAEC) gave us instructions that made us aware of the research project from the beginning. As a team, we would like to acknowledge the PAEC with our warm gratitude for guiding and

evaluating our work. Finally, we thank our friends and families for supporting us so far to reach the research objectives that are to be fulfilled in our research.

Table of Contents

Declaration 3

Acknowledgment 1

Table of Contents 1

List of Tables 1

List of Figures 1

Abbreviations 1

1. Introduction and Background 2

1.1 Project Scope 2

1.2 Motivation for the Study 3

1.3 Rationale and Justification 4

1.4 Research Problem 4

1.5 Research Objectives 5

Main Objective 5

Specific Objectives 6

1.6 Conceptual, Theoretical, and Practical Significance of the Research 7

1.6.1 Expected Outcomes 7

3. Methodology 1

4. Project Timeline 1

5. Experimental Plan/Project Design/System Design Overview 1

5.1 System Design Overview 1

5.2 System Architecture 2

5.3 Algorithm design 4

6. Results and Evaluation 1

6.1 Results of the Data Processing 1

6.2 Evaluation of Skill Extraction 2

6.3 Evaluation of the Matching System 3

6.4 Skill Gap Analysis 3

6.5 Overall System Analysis 4

7. Discussion 1

8. Conclusion 1

9. Limitations and Future Work 1

9.1 Limitations 1

9.2 Future Work 1

10. References 1

List of Tables

Table 1 : Project timeline 1

List of Figures

Figure 1 : Web scraping workflow 1

Figure 2 : Imgae before OCR preprocessing 2

Figure 3 : Image after preprocessing 2

Figure 4 : Skill Dictionary 3

Figure 5 : Cosine similarity 5

Figure 6 : streamlit 6

Figure 7 : Internship matcher 3

Figure 8 : Skill gap analysis 4

Abbreviations

Abbreviation	Full Form
OCR	Optical Character Recognition
NLP	Natural Language Processing
IT	Information Technology
UI	User Interface
API	Application Programming Interface
CSV	Comma-Separated Values
TF-IDF	Term Frequency – Inverse Document Frequency
ML	Machine Learning
AI	Artificial Intelligence
URL	Uniform Resource Locator
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
DB	Database
BOW	Bag of Words

Abstract

This research introduces a data-driven system for internship matching, which will help Sri Lankan students to find suitable internships based on their skill sets. With the rise of online job postings and the competitive nature of students, it is becoming increasingly difficult for students to find suitable internships. The suggested system utilizes web scraping, optical character recognition, natural language processing, and recommendation systems based on cosine similarity in order to automate the process. Internship advertisements have been gathered from the internet, including image-based job postings. The extracted text data is processed using a skill extraction pipeline based on dictionary matching and fuzzy string matching techniques. The system constructs skill vectors for both student profiles and internship descriptions, enabling similarity computation using cosine similarity. Based on this approach, the system recommends internships and identifies skill gaps for students. The results indicate that the system successfully identifies relevant internships and highlights in-demand skills such as web development technologies and cloud computing. This research demonstrates the practical applicability of NLP and recommendation systems in addressing real-world employment challenges in Sri Lanka.

1. Introduction and Background

In the present scenario, with the rapid evolution of technology, online job sites and company websites have become the main sources of internship opportunities. However, students face difficulties in searching for appropriate internships based on the high volume of job postings and the lack of structured information. In the case of Sri Lanka, internship postings are mostly available in the form of images or unstructured text, which makes it hard for students to obtain relevant information. Moreover, students do not have the means to assess their skills. In order to overcome this problem, this research suggests a system that can automatically extract internship opportunities and match the information with the student's skill set.

1.1 Project Scope

The scope of this research is limited to designing and developing a data-driven internship matching system for Information Technology-related internships in Sri Lanka. The aim of the system is to help students find appropriate internship opportunities by processing job advertisements and matching them with individual skills. The data-driven internship matching system will include several steps such as data collection, data processing, and recommendation. Data collection will include collecting internship data from online job portals and selected company career pages using web scraping. In addition, the system will include image-based job advertisement processing using Optical Character Recognition to extract text from image-based job posters, which are commonly used in Sri Lankan job platforms. The extracted data is then processed further through Natural Language Processing techniques to determine relevant technical skills. The predefined skill dictionary, along with fuzzy matching techniques, is employed to increase the precision level in skill extraction,

particularly in scenarios where data inconsistencies or OCR inaccuracies are involved. The matching feature has been developed based on a cosine similarity approach, in which both student profile and internship requirement data are represented as vectorized skill sets, enabling the system to recommend internships and determine missing skills for the student. The system has been developed to support internship recommendation, skill set identification, and missing skill identification for students, based on predefined skill dictionaries, fuzzy matching, and cosine similarity approaches. The system is developed as a prototype and does not include real-time integration with all job platforms and does not incorporate more sophisticated machine learning techniques like deep learning-based embeddings. However, the system provides a good foundation for scalable internship recommendation systems in Sri Lanka.

1.2 Motivation for the Study

The motivation behind this research is derived from the challenges faced by students in finding appropriate internship opportunities in Sri Lanka. The main problem is that the increasing number of job advertisements published in unstructured ways such as images and inconsistent text makes it difficult for students to analyze and determine their skills. Moreover, there is no automated system to recommend appropriate internship opportunities to students and to identify their skill gaps. Therefore, the motivation behind this research is to develop a data-driven system to facilitate students in their internship search and to make appropriate career decisions.

1.3 Rationale and Justification

This rapid development in digital job platforms has led to a huge number of internship opportunities available on these platforms. However, most of this data is unstructured and not easy to interpret. In Sri Lanka, most of the internship opportunities are posted in image or text formats that are not properly structured, making it difficult for students to obtain relevant data such as skills and qualification required for a particular internship. There is a gap in terms of skills between students and the industry, despite the availability of a number of internship opportunities. Most students use a manual approach in applying for internships, which is based on judgment. There is a lack of localized tools that can analyze job market trends and provide recommendations based on the skills of students.. This research is motivated by the need to address this gap with a data-centric approach. With the integration of web scraping, OCR, NLP, and cosine similarity-based matching, this system will provide an efficient and automated solution for internship discovery. Moreover, this system will not only provide relevant internship suggestions but will also identify gaps, thus helping students prepare for the job market. This research will contribute to the academic and practical aspects with the introduction of a scalable and localized internship matching system, particularly with the Sri Lankan context.

1.4 Research Problem

In Sri Lanka, students are facing problems in finding suitable internship opportunities due to unstructured and fragmented forms of job advertisements, where most internship

opportunities are published in image or text formats, making it harder to analyze the required skills from the advertisements. Hence, students are forced to use manual search techniques and judgment while applying for internships, which is causing inefficiency in matching students with industry needs, as no automated system exists to analyze advertisements, determine required skills, and suggest internships based on individual students' details. Moreover, no system exists to analyze advertisements, determine required skills, and suggest internships based on individual students' details, making it harder for students to apply for internships in an efficient manner. Additionally, the students may be unaware of the skills required for the current job market, which would affect their ability to adequately prepare for the internship. This difference between the skills of the students and the requirements of the job market underscores the need for a well-thought-out and systematic solution.

Hence, the research problem to be solved by the current study is:

How to design an automated system for extracting internship details and effectively matching the students with the internship programs while identifying the skill gaps?

1.5 Research Objectives

Main Objective

The main objective of this research is to develop a data-driven system that matches students with suitable internship opportunities based on their skill sets using cosine similarity techniques.

Specific Objectives

- To collect internship data from job portals and company career pages utilizing web scraping methods.
- To process the text data extracted from the image-based job advertisements utilizing Optical Character Recognition (OCR).
- To process the text data for further analysis.
- To recognize and extract the relevant technical skills utilizing Natural Language Processing (NLP) techniques, which include dictionary and fuzzy matching techniques.
- To represent the student profiles and internship requirements as vectorized skill sets.
- To design a cosine similarity-based matching algorithm to provide internship recommendations to the students.
- To recognize and highlight the skill gaps between the student profiles and internship requirements.
- To design a prototype system to provide internship recommendations and insights to the students.

1.6 Conceptual, Theoretical, and Practical Significance of the Research

This is significant because it solves a real-world problem faced by students in Sri Lanka when they are in search of internship opportunities. The proposed system will provide a structured and automated approach to analyze unstructured job advertisements and recommend personalized internship opportunities based on the skills of students. From a practical point of view, the proposed system will help students find appropriate internship opportunities more efficiently. It will help students identify the skills required by the current job market. It will help students improve their skills based on the gap between required and available skills. It will save students' time in searching for jobs. From an academic point of view, this research proves the usage of web scraping, Optical Character Recognition, Natural Language Processing, and cosine similarity in developing a recommendation system. The contribution of this research is that it introduces a localized solution, which is applicable to the Sri Lankan context. The Sri Lankan context is not commonly found in literature. In addition, this system can be extended in the future to include sophisticated machine learning techniques, different industries, and career guidance systems. Hence, this research is theoretically and practically significant in enhancing internship opportunities and employability for students.

1.6.1 Expected Outcomes

The expected result of this research would be the creation of a functional prototype system that would be able to successfully match students with internship opportunities according to their skill set. The system would be able to successfully extract internship opportunities from both text-based and image-based job ads and identify key technical skills using NLP

techniques. Moreover, the system would also be able to provide internship recommendations using cosine similarity and provide meaningful insights into gaps in skills for students and the industry. This would enable students to better understand the skills required for their desired roles. Moreover, this research would also be able to provide a structured set of internship opportunities and in-demand skills in the Sri Lankan IT job market. Overall, this system would be able to act as a functional tool for internship discovery and help students make better decisions in their careers

2. Literature Review

The rise of the use of digital platforms in the recruitment process has provided opportunities for the automation of internship discovery and recommendation. However, it has also presented challenges, including information overload, unstructured data, and the difficulty of relating student skills with industry requirements. This literature review explores existing literature on internship management systems, intelligent web scraping, natural language processing, skill extraction, and similarity-based recommendation techniques, with a focus on the research gap filled by this research. Internship is an essential component in the link between academic and professional life. However, the traditional system of internship matching is mostly inefficient and time-consuming. The research paper by Mydyti and Ware sheds light on the fact that the traditional system of internship management mostly relies on manual screening and decision-making, which may result in improper matching between students and organizations. The research paper suggests the integration of intelligent web scraping, machine learning, and natural language processing to enhance the efficiency of internship matching. However, the suggested framework is mostly theoretical and does not consider the practical difficulties of applying this system in a particular geographical location, i.e., Sri Lanka. Web scraping is a significant technique for collecting data on a large scale from different job portals and organizations' career sites. Previous studies have shown that web scraping is a technique for automating the collection of vacancies, job details, and employers' data from different sources on the Internet. In terms of internship management, web scraping is a technique for updating internship databases and reducing data collection efforts. Mydyti and Ware have emphasized that parsing HTML and APIs are major

techniques for collecting data on recruitment-related activities. At the same time, they have pointed out that there are certain technical difficulties in collecting data from different websites. In the current study, web scraping is a major technique for collecting data since internship advertisements are available on different platforms in Sri Lanka. In addition, they are not available in a single format. A major challenge in job advertisement analysis is that most of the data is unstructured. Natural Language Processing is often employed to analyze such data. Literature reviews on previous studies have shown that Natural Language Processing is effective in identifying key data such as skills, qualifications, and job roles from job advertisements. In internship matching systems, Natural Language Processing can be employed in feature extraction from internship posts and candidate data. Mydyti and Ware discuss Natural Language Processing in the context of feature extraction from job posts to obtain key data for improving the accuracy of recommendation systems. The study explains that text analysis can be employed to transform raw data from job vacancies into structured data fields for matching purposes. The idea is a strong justification for the current study, in which text data is extracted from advertisements for processing to identify technical skills and transform them into structured skill sets. Though NLP is a useful tool, the practical application becomes increasingly complex if the initial data is unclean or not uniformly formatted. This is especially relevant in the context of Sri Lanka, as many job postings are presented in the form of images, as opposed to text. In the research proposal, it is highlighted that job postings on sources such as TopJobs, relevant to the Sri Lankan context, present job postings in the form of posters, and thus, Optical Character Recognition must be performed. The proposal identifies this as a critical issue, relevant to the context of Sri Lanka, and highlights how existing global models do not address this issue adequately. Unlike many

existing research papers, this research will include OCR as an essential step, preceding NLP, for the extraction of skills. Another key concept that is discussed in the literature is that of skill gap analysis. The current labor market is characterized by the need for specific skills and professional competencies. Therefore, it is crucial for learning and training institutions to understand this need in order to bridge the gap. According to the project proposal, the technology sector in Sri Lanka faces a mismatch between the skills of graduates and the needs of the labor market. Also, there is no system to monitor the needs of the labor market in real time through the analysis of internship advertisements. The proposed research is crucial in bridging this gap through the extraction of frequently requested skills in the country's internship advertisements. The research is crucial to both students and learning and training institutions. Recommendation systems represent another significant domain of relevance. In existing works on recruitment/internship systems, there is a common tendency to use matching algorithms to evaluate and compare candidates and opportunities. These include collaborative filtering, content-based filtering, machine learning, and neural networks, among others. According to Mydyti and Ware, intelligent matching systems can enhance the quality of recommendations by fitting student profiles with internship opportunities based on data analysis. However, some of these systems were developed with complex infrastructures or large-scale training data in mind. In a localized system, a clearer and easier approach may be desired. Cosine similarity is a popular similarity metric used in information retrieval systems and recommendation systems. It has the advantage of being simple yet effective. In the context of the study, the cosine similarity metric can be used. Both student profile data and internship requirement data can be represented as skill vectors. Using the cosine similarity metric, the system can calculate the angle between the two vectors. Using the angle,

the system can calculate the degree of similarity between the student profile data and the internship requirement data. It can be particularly advantageous for the development of the system as a prototype, as the approach is mathematically sound, computationally efficient, and easy to understand. It also suits the goal of building a system that is not too complex. One of the significant themes that have been highlighted in the literature is the lack of context-aware solutions in the developing world. The research proposal explicitly highlights the localization gap, the linguistic complexity gap, the practical validation gap, and the emergent economy skill taxonomy gap. It is mentioned that the research that has been done so far is mostly geographically neutral and assumes the availability of standardized and high-quality data on the internet, but the internship data in Sri Lanka is in a fragmented state, is multilingual, and is of low quality. Thus, it is evident that although the theoretical basis for the development of intelligent internship management systems is available, the need still persists. From the reviewed literature, it can be established that previous works have established significant foundations in web scraping, NLP, and recommendation system development, but they have failed to fully address the problems in localized internship matching in Sri Lanka, as they were developed in a theoretical context or in environments with clean and easily accessible data. This research, therefore, addresses the real-world localized problem by integrating web scraping, OCR, text processing, skill extraction, and cosine similarity-based matching into one system, as opposed to previous works. Additionally, this research has generated a localized internship database and has identified in-demand technical skills for Sri Lankan students/interns. In other words, it can be summarized that the existing literature supports the need for intelligent data extraction and recommendation systems for internship systems, yet it also points to a research gap in terms

of implementation-based solutions. Thus, this research is important because it is not only a contribution to the existing theoretical research but also because it applies that research to practical use through the development of a system that is appropriate to the Sri Lankan internship system.

3. Methodology

This research employed a data-driven approach to a system development methodology to develop a localized internship matching system for students in Sri Lanka. The data-driven approach to a system development methodology includes data collection, text extraction, preprocessing, skill extraction, data preparation, similarity-based matching, and prototype development. The entire process aims to transform unstructured internship advertisements into a structured skill-based form and match them against student data to provide recommendations for internships. The first step of the methodology is data collection. The internship advertisements were collected from different Sri Lankan job websites and companies' career pages through web scraping.

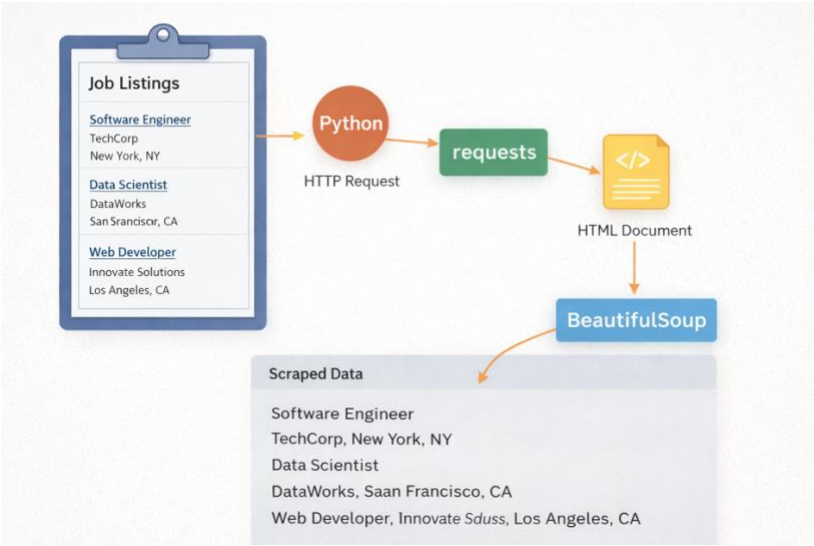

Figure 1 : Web scraping workflow

Most of the advertisements on the Sri Lankan job websites were in the form of images or posters. Therefore, the data collection step also included downloading the images of the job

posters and collecting the available information such as title, company name, and advertisement links. The second part of the process was text extraction. In terms of image-based advertisements, Optical Character Recognition (OCR) was employed to obtain raw text data from job posters. This is a significant part of the overall methodology since a substantial number of internship advertisements in Sri Lanka are in poster form. OCR allowed for the transformation of such image-based data into text form.

Figure 2 : Imgae before OCR preprocessing

Figure 3 : Image after preprocessing

Nevertheless, due to the presence of noise or misspellings in the text obtained from OCR, further processing was required. The third stage involved text preprocessing and normalization. In this case, the extracted raw texts were subjected to cleaning, where the texts were converted to lowercase, symbols were removed, whitespaces were normalized, and common OCR-related corrections were made. Additionally, the normalization of skill name and aliases was carried out. In other words, variations of the same word, such as “node js” and “node.js,” were normalized. The fourth stage was skill extraction using NLP-based techniques. In this stage, a hybrid approach was used by combining a predefined skill dictionary, alias mapping, and fuzzy string matching.

```
SKILL_CATEGORIES = {

# ================ PROGRAMMING ================
"python":"programming","java":"programming","javascript":"programming","typescript":"programming","c":"pro
"rust":"programming","kotlin":"programming","swift":"programming","r":"programming","scala":"programming",
"dart":"programming","matlab":"programming","bash":"programming","shell":"programming",

# ================ WEB ================
"html":"web","css":"web","sass":"web","bootstrap":"web","tailwind":"web",
"react":"web","angular":"web","vue":"web","node.js":"web","express":"web",
"next.js":"web","nuxt":"web","jquery":"web","ajax":"web","rest api":"web",
"graphql":"web","json":"web","xml":"web","frontend":"web","backend":"web","full stack":"web",

# ================ DATABASE ================
"sql":"database","mysql":"database","postgresql":"database","postgres":"database",
"sql server":"database","mssql":"database","oracle":"database","mongodb":"database",
"firebase":"database","redis":"database","database":"database",

# ================ DATA / AI ================
"pandas":"data","numpy":"data","matplotlib":"data","seaborn":"data",
"scikit-learn":"data","tensorflow":"data","keras":"data","pytorch":"data",
"machine learning":"data","deep learning":"data","data science":"data",
"data analysis":"data","data analytics":"data","nlp":"data",
"power bi":"data","tableau":"data","excel":"data",

# ================ CLOUD ================
"aws":"cloud","azure":"cloud","gcp":"cloud","docker":"cloud","kubernetes":"cloud",
"terraform":"cloud","jenkins":"cloud","ci/cd":"cloud","linux":"cloud",
"devops":"cloud","cloud computing":"cloud",

# ================ MOBILE ================
"android":"mobile","ios":"mobile","flutter":"mobile","react native":"mobile",

# ================ TESTING ================
"selenium":"testing","pytest":"testing","junit":"testing",
"testing":"testing","qa":"testing","quality assurance":"testing","postman":"testing",

# ================ TOOLS ================
"git":"tools","github":"tools","gitlab":"tools","bitbucket":"tools",
"jira":"tools","figma":"tools","wordpress":"tools","shopify":"tools",
"erp":"tools","sap":"tools","ui/ux":"tools","photoshop":"tools",

# ================ SECURITY ================
"networking":"security","cybersecurity":"security","information security":"security",
```

Figure 4 : Skill Dictionary

The skill dictionary used was comprised of technical skills related to the domain of the IT internship, including programming languages, web development, database management, cloud computing, and security-related concepts. First, exact matching was used to identify the direct occurrence of skills in the text. Furthermore, fuzzy matching was used to identify skills that were impacted by noise during OCR or slight changes in the text. This allowed for more accurate identification of relevant technical skills from unstructured job descriptions. After the skill set was extracted, the next stage was the structured dataset generation stage, which is the fifth stage in the process. The internship advertisements were represented as a record in the dataset, and the record had elements like job titles, companies, locations, technical skills, and the URLs of the advertisements. The next stage is the generation of the processed internship dataset, which is useful in the analysis and recommendation processes. The filtering process in this stage reduced the non-relevant and low-quality records in the dataset, making the final dataset more relevant in terms of technology internship opportunities. The sixth stage was student profile representation and vectorization. Student profiles were represented as lists of self-declared technical skills. Student skill sets and internship skill sets were both converted into vector representations using the bag of words approach. Count Vectorizer was used to transform the extracted skills into numerical vector representations. In this representation, each feature represented a unique skill term, and comparisons could be made between student profiles and internship requirements.

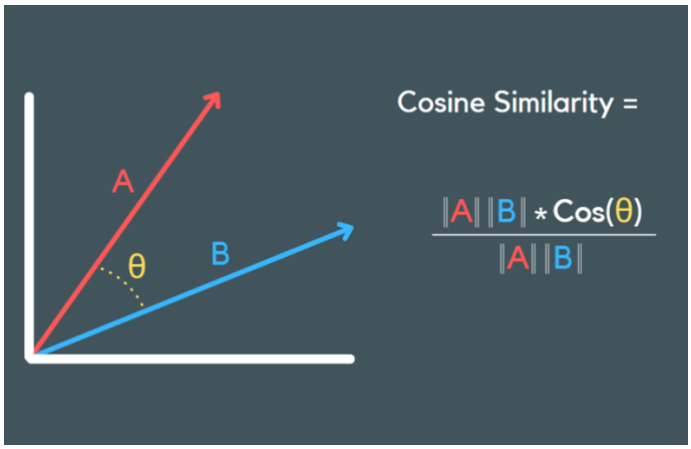

Figure 5 : Cosine similarity

The seventh stage was that of matching, and this was achieved through the application of the matching algorithm, specifically cosine similarity. Cosine similarity was used because it is efficient and easily understandable in measuring similarity between two vectors. In this research, one vector was for the student's skills and the other for the extracted requirements for an internship. The score was then generated for the two vectors using cosine similarity to ascertain how well they matched. The score was used to rank the internship opportunities and provide the most relevant results to the user. In addition, the system had the ability to carry out skill gap identification. After the matching process, the system would compare the skills required from the extracted skills from the internship advertisement. The system would then compare the student's skill set with the required skills from the internship advertisement. This aspect of the system had practical implications, as it helped the student identify the skills he or she needed for the particular internship. The last step in the methodology was the development of the prototypes. A web application was developed using Streamlit, and it was used to show the capabilities of the suggested system.

Figure 6 : streamlit

The users were able to enter their technical skills, view the suggested internships, and see the percentage matches and missing skills. The system also provided users with some market information based on the analyzed internship data. This was a proof of concept for the suggested system for localized internship matching. Overall, this methodology employed web scraping, OCR, NLP, and cosine similarity in an integrated manner. This was an appropriate methodology for this Sri Lankan internship domain because it effectively tackled the problems with unstructured job advertisements, skill extraction, and internship recommendations for students.

4. Project Timeline

Table 1 : Project timeline

Phase	Week 1-4	Week 4-8	Week 8-12	Week 12-16	Week 16-20	Week 20-24
Proposal and literature review						
Scraper development and data collection						
NLP pipeline and analysis						
Prototype development						
Stakeholder validation						
Writing and finalization						

5. Experimental Plan/Project Design/System Design Overview

5.1 System Design Overview

The proposed system has been designed as a modular and data-driven system, which brings together various modules for processing internship-related data and generating recommendations. The system has been designed as a pipeline-based system, where every module has been designed for a specific task. The system has been designed as a three-component system: data acquisition, data processing, and recommendation generation. These three components of the system help in processing raw internship advertisements and generating meaningful internships for students. The data acquisition component is responsible for the retrieval of internship advertisements from various sources. These sources may include job portals and company career pages. Web scraping is used to retrieve the available job data. Image-based advertisements are also downloaded for further processing. The data acquisition component ensures the retrieval of diverse internship data from the Sri Lankan job market. The data processing part includes the conversion of unstructured data into a structured format. This includes the use of OCR techniques to extract text from advertisements in image format, followed by text pre-processing and normalization. Further, the text is processed by the skill extraction module, which uses a pre-defined skill dictionary to identify the skills required in the text. Finally, the dataset in a clean and structured format is obtained, including information related to internships along with the skills required. The recommendation part of the system is responsible for matching student profiles with internship opportunities. The skills of students are collected and represented as vectors. In

the same manner, skill sets of internships are represented as vectors by using the bag of words model. Cosine similarity is then applied on the vectors of student skills and internship requirement skills. Based on similarity scores, internships are ranked and presented to users. Besides recommendation, the system has a skill gap analysis tool that compares the skills of the student with the requirements of the internship. This tool offers the student valuable insights on how to enhance their employability skills. The entire system is implemented as a prototype, and it is web-based, enabling users to interact with the system easily. The modularity of the system ensures that any part of it can be enhanced for future purposes, such as real-time data updates, sophisticated machine learning algorithms, and inclusion of more job portals. In summary, the proposed system design ensures an efficient flow from data collection to recommendation, and it is a practical and efficient solution for internship matching.

5.2 System Architecture

The proposed framework for the internship matching system is developed in a modular fashion, incorporating data acquisition, processing, and recommendation aspects in a cohesive manner. This architecture facilitates a systematic transformation of unstructured internship advertisements into a structured form for generating personalized recommendations for the students.

The system architecture consists of five main layers: **Data Collection Layer, Text Extraction Layer, Data Processing Layer, Matching Layer, and Presentation Layer**.

The Data Collection Layer is in charge of retrieving internship data from various online

sources, including job sites and company career sites. Web scraping methods are used to retrieve available job data such as job titles, companies, and ad links. Additionally, job posting images are fetched as they will be used in sites where job postings are image-based. The Data Collection Layer provides data availability for further processing. The Text Extraction Layer is designed to convert the visual data into text. In the text extraction layer, Optical Character Recognition (OCR) is used to extract the text from the job advertisements. The text may contain noise like incorrect characters. Therefore, the output of the text extraction layer is treated as raw. The Data Processing Layer includes text cleaning, normalization, and skill extraction. During this phase, the text is cleaned and preprocessed by removing unwanted symbols, correcting OCR, and normalizing text. The hybrid Natural Language Processing (NLP) technique is used for extracting skills from text, where a predefined dictionary and fuzzy matching algorithms are used. The processed data is stored as a dataset containing internship details and skill sets. The Matching Layer is the central part of the system. It models both student profiles and internship requirements as sets of skills represented as vectors, using a bag of words model. Cosine similarity is then used to compute the similarity of these vectors. Based on similarity scores, internships are ranked in descending order of their relevance. This layer also provides skill gap analysis by determining skills required by internships and absent in the student profile. The Presentation Layer offers an interface for user interaction. A web-based prototype will be developed to allow students to input their skills and view the recommended internships. This layer ensures that the system outputs are presented in a manner that can be easily understood by the end-users. This modular structure enables each of the layers to perform independently of each other while at the same time promoting a seamless flow of information throughout the system. This

structure promotes flexibility and scalability, thus paving the way for future enhancements such as the incorporation of advanced machine learning techniques, dynamic data update, and expansion to other domains outside IT internships. The system architecture has been designed in a manner that promotes efficiency in the management of unstructured internship information and the generation of personalized recommendations, thus making it suitable for addressing the challenges of internship matching in Sri Lanka.

5.3 Algorithm design

The basic idea of the proposed system is that it will be based on a similarity-based matching algorithm. This algorithm will be responsible for comparing the skill set of the students with the internship requirements. The algorithm is designed in such a manner that it will be able to convert the text data of the skills into numerical form and calculate the similarity score using the cosine similarity measure. This will allow the proposed system to rank the internship opportunities in accordance with their relevance to the skill set of the students. The first step of the algorithm is the representation of the skills. In the proposed algorithm, the student profile and the internship descriptions will be represented in the form of a set of technical skills. These skills will be extracted and normalized during the data processing stage. A unified set of all skills will be created in order to facilitate their mathematical comparison. The second step is vectorization. In this case, a bag-of-words approach is used to create a numerical vector out of the skill sets. This vector represents the presence or absence of a certain skill in a given vocabulary. In relation to a certain student or internship, a vector represents a set of binary or frequency-based values related to a certain skill. This process ensures that both student profiles and internship requirements are represented in a

common vector space.

Assuming that the student skill vector is represented as A and the internship skill vector as B. Each dimension of these vectors represents a certain skill in a global vocabulary.

The third step in the process is computing the similarity using a cosine similarity approach. Cosine similarity measures the cosine of the angle between two vectors to determine how well they are aligned. It can be computed as follows:

$$similarity(A,B) = \frac{A \cdot B}{\|A\| \times \|B\|} = \frac{\sum_{i=1}^{n} A_i \times B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \times \sqrt{\sum_{i=1}^{n} B_i^2}}$$

where :

- A·B is the dot product of the two vectors
- ||A|| and ||B|| are the magnitudes of the vectors

The similarity score thus obtained will vary between 0 and 1, where a score closer to 1 indicates a higher degree of similarity between the skills of the student and the requirements of the internship. The fourth step in the process is the ranking and recommendation step. In this step, the similarity score of each student's profile is calculated against all the internship records in the database. The internships are then arranged in a list in descending order based on their respective similarity scores. Apart from the recommendation, the algorithm also carries out the skill gap analysis. It is done by matching the skills required by the internship

against the skills possessed by the student. The skills that are not possessed by the student are shown as a gap. It is beneficial to the student since it gives them the necessary insights to enhance their profile. The choice of the algorithm is based on its simplicity, efficiency, and interpretability. Unlike complex machine learning algorithms, cosine similarity does not require large datasets for training and can deliver accurate results using relatively small datasets. Also, the vector space representation makes it easy to extend the proposed model, for instance, by using weighting methods such as TF-IDF and moving on to more sophisticated methods later on. In summary, the proposed algorithm is an efficient and computationally lightweight internship matching solution, and it is suitable for use in a prototype as proposed in this research.

6. Results and Evaluation

This chapter discusses the results obtained from the proposed internship matching system and assesses the performance of the proposed approach in data extraction, skill recognition, and recommendation accuracy. The proposed approach is evaluated based on the dataset collected from Sri Lankan job portals and company career pages, containing around 5000 advertisements for internships.

6.1 Results of the Data Processing

The data collection and data preprocessing phases were successful in transforming the unstructured internship advertisements into a structured data set. The system was able to extract text data from text-based and image-based job posts using web scraping and OCR techniques. After data preprocessing and normalization, relevant internship data was filtered from the data set based on IT-related internship opportunities.

The skill extraction module was able to identify a set of commonly requested skills. The most frequent skills in the data set after processing are:

- CSS
- AWS
- HTML
- JavaScript

- PHP
- Python
- Java

These skills suggest that web development skills and cloud computing skills are in high demand in the Sri Lankan internship market.

6.2 Evaluation of Skill Extraction

The effectiveness of the skill extraction was evaluated qualitatively by matching the extracted skills with the original job description. Using the hybrid approach of a predefined skill dictionary and fuzzy matching techniques, the detection of skills was improved, even in the event of OCR errors and text inconsistencies. For example, inconsistencies such as misspellings and formatting of terms, such as “pyton” rather than “python” or “node js” rather than “node.js,” were successfully identified and normalized. This indicates that the use of the fuzzy matching techniques improved the effectiveness of the extraction process. However, certain limitations have been observed:

- Generic words like “testing” or “database” have been extracted without proper context
- Rare technologies have not been captured due to the limitations of the skill dictionary

Despite the above limitations, the accuracy of the extraction process is satisfactory for the

purpose of internship matching.

6.3 Evaluation of the Matching System

The performance of the internship matching system was assessed by testing it with a set of student skill profiles. Recommendations for internships were provided in a ranked manner based on the scores obtained from the cosine similarity measure. For instance, a student profile with skills such as “Python, HTML, CSS” was considered for testing the system. Internships that require similar technologies and higher similarity scores were recommended. Internships that require dissimilar skills are provided with low similarity scores and ranked accordingly. The use of the cosine similarity measure was effective in assessing the relevance between a student profile and an internship. The ranking mechanism ensured that the best internship is always placed at the top of the recommendation list.

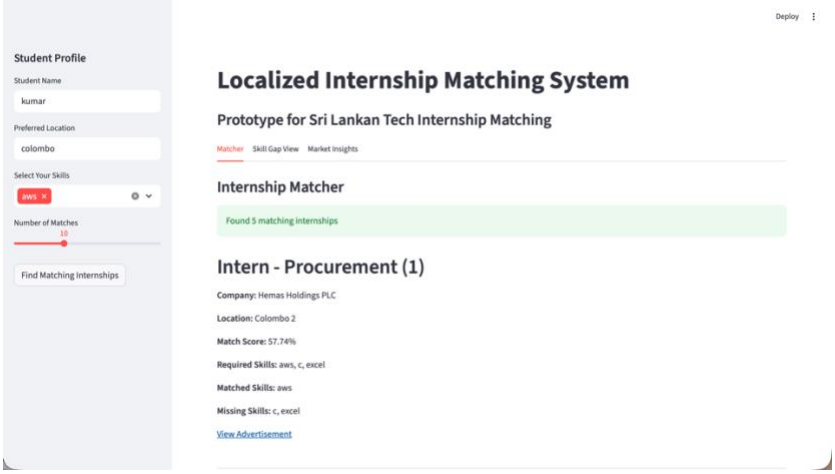

Figure 7 : Internship matcher

6.4 Skill Gap Analysis

Another result of the system is the recognition of skill sets. The system compared the skills

required for each recommended internship and the skill set of the student. It identified the missing skills for the student. For example, for a recommended internship that requires “HTML, CSS, JavaScript,” and the student has a skill set of “HTML and CSS,” the system identified “JavaScript” as a missing skill. This is useful for the student because it helps them understand their weaknesses.

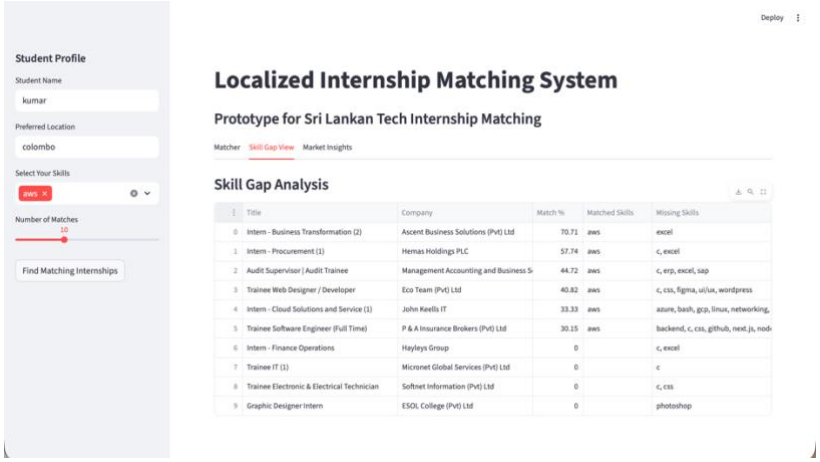

Figure 8 : Skill gap analysis

6.5 Overall System Analysis

Overall, it was demonstrated that the system is able to:

- Successfully collect and process internship data from various sources
- Extract relevant technical skills from unstructured text
- Generate meaningful internship recommendations using cosine similarity
- Identify skill gaps for students

From this, it is evident that the proposed approach is effective for creating a prototype

internship matching system. Although it is not optimized, it is a strong base for further improvements and optimizations.

7. Discussion

The results obtained through the present study show that the proposed method for an internship matching system can efficiently transform the job advertisements posted in an unstructured format into useful recommendations for students. The application of the cosine similarity algorithm was found to be an effective tool for efficiently matching student profiles with the required skills for an internship position. This algorithm was successfully used for ranking the available internships based on their relevance, and the results showed that the positions requiring similar skills were always ranked first, indicating that the proposed method for calculating the similarity between skills was effective for the purpose of recommendation. The performance of the skill extraction component was also reasonable despite the difficulties in dealing with noisy data from OCR and inconsistent formatting in job advertisements. The integration of a predefined dictionary and fuzzy matching techniques has significantly enhanced the system's performance in recognizing skills despite minor discrepancies. However, some limitations were identified in terms of extracting general words without sufficient context and failing to identify new technologies that were not included in the predefined dictionary. Adding a feature of skill gap analysis provides practical value for the system. Unlike conventional methods of searching for jobs, this system does not only help students find internship opportunities but also provides insights into skills that they lack. From a broader viewpoint, the results emphasize the significance of data-oriented methods for resolving employment-related issues. In particular, the system shows the efficacy of even a simple and comprehensible method like cosine similarity when combined with good data preprocessing and feature extraction. The current limitations of the proposed

system include issues related to the size of the dataset, the accuracy of the OCR tool, and the size of the skill dictionary. This demonstrates that although the proposed system is effective, it still requires further improvements for deployment. The study has proven the feasibility and usefulness of the proposed method for the Sri Lankan scenario and established a foundation for further enhancements for an intelligent internship recommendation system.

8. Conclusion

The research successfully proposed a data-driven internship matching system that aims to help Sri Lankan students find suitable internship opportunities according to their skills. The proposed system addressed the issues faced by unstructured job advertisements by integrating web scraping, Optical Character Recognition, Natural Language Processing, and Cosine Similarity-based Recommendation into one workflow. The proposed internship matching system showed its potential to gather internship-related data from various sources, extract useful information from unstructured text and image-based job advertisements, and recognize relevant technical skills using a hybrid extraction method. By modeling both student and internship requirements as a set of vectorized skill sets, the proposed system successfully applied Cosine Similarity to recommend internship opportunities to students. In addition, the system helped to gain valuable insights by using skill gap analysis, where students could comprehend the skills needed by the industry and enhance their employability. The findings from the study suggest that web development and cloud-related skills are highly required within the internship market of Sri Lanka, and hence there is a need to focus on matching the skills of students with the current trends. Although the proposed system is considered a prototype and has some limitations, such as inaccuracies in OCR and a predefined dictionary, it is considered a robust base for future developments. The research proves that even a simple and understandable similarity-based approach can lead to significant and useful outcomes by using efficient processing methods. Finally, in conclusion, the contribution of this research can be considered academic and practical in terms of introducing a localized solution for an internship matching system in the context of Sri Lanka. The proposed system has the

potential to be improved and implemented as a platform in the future.

9. Limitations and Future Work

9.1 Limitations

Although the proposed system of internship matching has shown its functionality as a prototype, there are some limitations of the system that were identified during the research. One of the major limitations of the system would be the usage of Optical Character Recognition (OCR) in recognizing the text from the image-based job advertisements. There might be some errors in the recognized text, such as spelling errors, which could affect the accuracy of the skills being extracted. Another limitation of the proposed model is the predefined skill dictionary used for the model. Even though the dictionary contains many technical skills, it may not contain all the new technologies available. Therefore, the model may not be able to detect all the skills used in the advertisements. The dataset used for the proposed model has many limitations too. Even though a large number of advertisements were used for the model, they may not cover all the available internship opportunities available in Sri Lanka. In addition, the proposed model is mainly used for IT-related internships only. Furthermore, in the cosine similarity approach, though it is simple to interpret, it does not take into consideration any contextual meaning or importance (weights) of any skill. It treats all skills equally; that is, they are given equal importance. This might affect the accuracy of the recommendation.

9.2 Future Work

There are certain improvements that can be made to further enhance the system in future studies. Firstly, the accuracy of text data can be improved by using more advanced models

in OCR or image preprocessing techniques. This can further enhance the accuracy of the data. Secondly, the skill extraction process can be further improved by using dynamic skill dictionaries that can be updated automatically. Machine learning or deep learning-based NLP can be used to get a better understanding of the context to extract skills more accurately. Thirdly, the matching algorithm can be improved by incorporating different weighting schemes such as the use of TF-IDF or by incorporating advanced recommendation systems. This would allow the system to assign different weights of importance to skills. Moreover, the system can be extended to include the ability to collect data in real-time from various job platforms and company career pages. This would allow the system to provide up-to-date recommendations for internships. Lastly, the prototype can be built into a full-scale application, including a user-friendly interface, user accounts, and integration into academic systems. The application can be extended beyond IT internships to include other domains, thereby making it a comprehensive career guidance application.

10. References

[1] C. D. Manning, P. Raghavan, and H. Schütze, Introduction to Information Retrieval. Cambridge, U.K.: Cambridge University Press, 2008.

[2] D. Jurafsky and J. H. Martin, Speech and Language Processing, 2nd ed. Upper Saddle River, NJ, USA: Pearson, 2009.

[3] S. Mydyti and M. Ware, "Integrating Intelligent Web Scraping Techniques in Internship Management Systems," 2023.

[4] A. Rajaraman and J. D. Ullman, Mining of Massive Datasets. Cambridge, U.K.: Cambridge University Press, 2011.

[5] G. Salton and M. J. McGill, Introduction to Modern Information Retrieval. New York, NY, USA: McGraw-Hill, 1983.

[6] Tesseract OCR Engine, "Tesseract Open Source OCR Engine," 2024. Available: https://github.com/tesseract-ocr/tesseract

[7] RapidFuzz Library, "RapidFuzz: Fast String Matching in Python," 2024. Available: https://github.com/maxbachmann/RapidFuzz

[8] Scikit-learn Developers, "Scikit-learn: Machine Learning in Python," 2024. Available: https://scikit-learn.org

[9] TopJobs Sri Lanka, “TopJobs Job Portal,” 2024. Available: https://www.topjobs.lk

[10] WSO2, “WSO2 Careers,” 2024. Available: https://wso2.com/careers

[11] Virtusa, “Virtusa Careers,” 2024. Available: https://www.virtusa.com/careers

[12] IFS, “IFS Careers,” 2024. Available: https://www.ifs.com/en/about/careers